NYMEX HEATING OIL / ULSD FUTURES (HO)

Spread & Curve Structure — Fundamental Reference

Distillate Markets, Diesel, Heating Oil, Marine Fuel, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of NYMEX Heating Oil / Ultra-Low Sulphur Diesel futures (ticker: HO), contract size 42,000 US gallons (1,000 barrels), quoted in USD per gallon and delivered at New York Harbor. The contract was formally renamed "ULSD" in 2013 when ultra-low sulphur diesel replaced heating oil as the primary deliverable grade, but it continues to be known colloquially as "Heating Oil" or "HO." The contract is simultaneously the global benchmark for diesel and distillate products and the domestic heating fuel reference for the US Northeast — two demand profiles with markedly different seasonal structures. This duality, combined with the contract's role as the global gasoil benchmark through the ULSD-ICE Gasoil relationship, makes HO one of the most informationally rich petroleum spread markets. Understanding the refinery economics, the dual demand structure, regulatory specifications, and the global trade flows that connect this market to European, Asian, and Middle Eastern distillate markets is the foundation of spread and butterfly trading on the HO curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

NYMEX HO is physically delivered at New York Harbor — the same delivery point as RBOB gasoline. This shared delivery location means that HO and RBOB are directly comparable at the point of delivery, and their relative pricing reflects both refinery yield decisions and end-user demand seasonality.

Parameter	Detail
Exchange	CME Group — NYMEX
Ticker	HO (front month)
Common names	Heating Oil futures, ULSD futures, NY Harbor ULSD. Informally: "Heating Oil" despite the ULSD rename in 2013.
Contract size	42,000 US gallons (1,000 barrels)
Quotation	USD per gallon; minimum tick = $0.0001/gallon = $4.20 per contract
Contract months	All 12 calendar months; active out to approximately 3 years. Liquid concentration in front 6–12 months; winter months carry additional liquidity from heating demand hedging.
Delivery grade	Ultra-Low Sulphur Diesel (ULSD): maximum 15 ppm sulphur. Must meet ASTM D975 (No. 2 diesel) or D396 (No. 2 heating oil) specifications. Colour requirement (dye-free for highway diesel; red-dyed for heating oil) managed at terminal level, not futures contract level.
Settlement type	Physical delivery via pipeline or barge at New York Harbor, primarily through Buckeye Pipe Line, Colonial Pipeline terminus, or marine facilities.
Last trading day	Last business day of the month preceding the delivery month
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour); liquid hours approximately 07:00–14:30 ET
DV01	$0.0001/gallon = $4.20 per contract
Contract value at $3/gal	$126,000 per contract
Relationship to ICE Gasoil	ICE Gasoil (GO, denominated in USD/MT at Rotterdam) is the European equivalent. The HO-Gasoil spread ("the arb") drives transatlantic distillate trade flows and is a key variable for prompt HO spreads.

1.1 The ULSD Rename and What It Changed

- Until 2013, the contract primarily tracked No. 2 heating oil (distillate fuel used for residential and commercial heating). The 2013 delivery grade change to ULSD (max 15 ppm sulphur) reflected the completion of the EPA's transition to ULSD as the standard for all on-road diesel fuel in the US, making the old high-sulphur heating oil non-deliverable.
- The practical consequence: the contract now prices ultra-low sulphur distillate at NYH — a product that is interchangeable between highway diesel, home heating oil, jet fuel blending, agricultural diesel, and marine gasoil (with limited sulphur tolerance). This versatility is a key feature of the distillate pool: the same barrel can serve multiple end-uses, and the market allocates it based on relative prices.
- Heating oil for residential use is still the same product (No. 2 distillate, ULSD-compliant), just dyed red at the terminal to indicate it is for off-road use (tax purposes). The futures contract does not carry the dye — physical receivers add it at the terminal.

1.2 The HO Crack Spread

- **Simple HO crack**: HO price ($/gallon × 42 gallons/barrel) minus WTI crude price ($/barrel). Represents the gross refinery margin for producing distillate.
- **3-2-1 crack spread**: (2 × RBOB + 1 × HO − 3 × WTI) / 3. Combined measure of overall refinery margin. HO and RBOB crack spreads must be read together because refineries co-produce both products and optimise the yield mix based on relative crack economics.
- **HO typically trades at a premium to RBOB in winter** (heating demand) and at a discount in summer (gasoline driving season dominates). This seasonal relationship is one of the most consistently observed patterns in petroleum markets.

2. The Distillate Pool — Multiple End Uses, One Contract

The single most distinctive feature of the HO market is that the underlying commodity is fungible across multiple end-use markets. Understanding where distillate goes and how demand shifts between uses is the starting point for all supply/demand analysis.

End Use	US Consumption (approx.)	Seasonal Pattern	Key Driver	HO Curve Sensitivity
On-road highway diesel (trucks, buses)	~2.8–3.2 mb/d	Moderate seasonality — higher in Q4 (harvest, pre-holiday freight)	GDP, freight volumes, truck miles travelled, e-commerce growth, diesel engine fleet size	Baseline year-round demand. Most relevant for mid-curve (C2–C6). ISM and freight data are key signals.
Residential heating oil (primarily Northeast US)	~0.3–0.4 mb/d	Strongly winter-peaked. Near-zero in April–September.	Temperature (HDD), PADD 1 heating oil stocks, population in oil-heated homes (declining), alternative heat source competition	Primary driver of winter (Nov–Feb) HO premium and backwardation. PADD 1 HO stock levels are the key prompt signal.
Agricultural diesel (tractors, irrigation)	~0.2–0.3 mb/d	Spring planting (Apr–May) and autumn harvest (Sep–Nov) peaks	Planted acreage, crop calendar, weather (planting delays)	Creates secondary seasonal peaks in spring and autumn spreads.
Rail (locomotive fuel)	~0.1 mb/d	Relatively flat seasonally	Rail traffic volumes, intermodal competition	Stable baseline, low spread sensitivity on its own.
Marine fuel (bunkers)	~0.15–0.2 mb/d domestic; larger global market	Moderate, tied to shipping activity	Global trade volumes, IMO sulphur regulations, shipping rates (Baltic indices)	IMO 2020 VLSFO demand shifted some bunker demand from HSFO to distillate blends — medium-term structural increase in ULSD-compatible marine demand.
Jet fuel / kerosene (a separate spec)	~1.5–1.7 mb/d	Summer peak (aviation travel), winter aviation demand also high	Passenger and cargo traffic, airline capacity, load factors	Jet fuel is a close substitute in refinery yield terms. When jet-HO spread is wide, refinery runs can be adjusted. Jet demand affects the broader distillate pool level.
Power generation (emergency / peaking)	Relatively small	Winter peak (backup generation during cold snaps)	Natural gas price spikes, extreme cold, grid reliability	A natural gas price spike or grid emergency can temporarily increase distillate burn for power — a bullish tail risk for prompt HO.

The fungibility of the distillate pool is both its strength (demand from multiple sectors provides a floor) and a complication for analysis (a change in one end-use demand can be partially offset by shifts in another, damping volatility). The market allocates barrels by

price: when heating oil demand is high, its price rises relative to diesel, which pulls supply from the diesel pool; the two uses compete for the same barrel.

3. Heating Oil Demand — The US Northeast

Residential and commercial heating oil demand in the US Northeast (primarily New England and New York) is the seasonal demand driver that creates the winter premium embedded in the November through February HO contracts. Understanding this market's structure and secular trends is essential.

3.1 The Northeast Heating Oil Market — Structure

- **Market size**: approximately 4–5 million homes in the Northeast US still use heating oil as their primary heat source, concentrated in Connecticut, Maine, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont. Total residential heating oil consumption is approximately 0.3–0.4 mb/d during the winter season.
- **Secular decline in heating oil use**: the number of oil-heated homes has declined consistently for decades as natural gas conversion, heat pump installation, and other alternatives expand. This is a slow-moving structural bearish signal for the winter HO premium over multi-year time horizons — but year-to-year, the dominant variable is still the weather.
- **Dealer inventory structure**: heating oil is delivered to homes by oil dealers who carry their own inventory in small tank farms (often 20,000–500,000 gallon capacity). Dealers pre-buy supply (sometimes using futures hedges) before winter to lock in costs for their customers. This pre-buying behaviour creates demand for the October–November HO contracts during summer months, contributing to the seasonal premium.
- **Automatic delivery vs will-call**: most residential heating oil customers are on "automatic delivery" contracts — the dealer monitors consumption via degree-day calculations and delivers without the customer's active instruction. This makes residential heating oil demand more weather-correlated and less price-elastic than other energy demand categories.

3.2 Heating Degree Days and Heating Oil Demand

- **HDD-demand relationship**: residential heating oil consumption is approximately linearly correlated with population-weighted HDDs in the Northeast. The EIA publishes regional HDD data weekly, and commercial weather services provide forecast HDDs for the coming 1–3 weeks. A week with 300 HDD in New England is roughly twice the heating demand of a 150 HDD week.
- **The temperature vs PADD 1 stock interaction**: the most important fundamental framework for the winter HO spread is the combination of temperature forecasts and PADD 1 distillate inventory levels. Cold forecasts with low stocks = bullish (potential shortage). Mild forecasts with high stocks = bearish (excess supply). Cold forecasts with high stocks = neutral to mild bullish. Mild forecasts with low stocks = depends on duration of mildness.
- **NOAA 30-day and 90-day temperature outlooks**: the Climate Prediction Center's monthly and seasonal temperature outlooks are closely watched in October–November as the winter heating season approaches. A forecast for a colder-than-normal winter across the Northeast creates a structural bid for the November–February HO strip before the season begins.

3.3 The Northeast Energy Security Issue

- New England is the most energy-supply-constrained region in the continental US: it has minimal natural gas pipeline capacity additions since the 1990s (due to permitting opposition), heavy reliance on imported LNG for winter gas supply, and high electricity prices due to grid composition. This infrastructure constraint means that during extreme cold events, the Northeast simultaneously faces:
- Natural gas pipeline capacity constraints (Algonquin pipeline constrained → gas prices spike at Citygates), which shifts some heating demand from gas to oil (dual-fuel buildings switching to oil backup), creating a simultaneous demand increase for both natural gas (pipeline) and heating oil (from fuel switching). This dual-fuel switching mechanism creates a correlated demand spike across

both markets.

- Dual-fuel buildings (primarily large commercial and residential buildings that can burn either gas or oil) switch from gas to oil when gas prices spike above a threshold (typically Algonquin Citygates above $20–30/MMBtu). These events are acute and concentrated in the coldest winter weeks.

4. Highway Diesel — The Structural Demand Base

While heating oil creates the seasonal demand spikes, highway diesel is the structural demand base that determines the mid-curve level of HO futures. Diesel demand is tied to economic activity, freight volumes, and the composition of the vehicle fleet.

4.1 US Diesel Demand Drivers

- **Truck freight volumes**: Class 8 heavy-duty trucks (semis, 18-wheelers) are the largest single user of highway diesel. The American Trucking Associations (ATA) Truck Tonnage Index is a monthly measure of freight volume and a leading indicator of diesel demand. The Cass Freight Index and DAT Freight Intelligence provide additional high-frequency freight demand signals.
- **E-commerce growth**: the expansion of last-mile delivery (driven by Amazon, UPS, FedEx, and other carriers) has increased light-to-medium duty diesel truck demand significantly since 2015. This demand is less cyclical than heavy freight because consumers continue e-commerce purchasing even in recessions.
- **GDP and industrial production**: diesel demand correlates strongly with GDP growth because freight volume is a function of the volume of goods produced, distributed, and sold. ISM Manufacturing PMI (above/below 50) is a leading indicator of industrial freight and therefore diesel demand.
- **Diesel engine displacement by EVs**: electric truck adoption (Class 8) is nascent but represents a long-term structural bearish signal for diesel demand. The penetration rate is slow — diesel trucks have a working life of 10–15 years, and commercial EV economics are still maturing. Near-term (1–5 year) HO spreads are not materially affected.
- **Agricultural diesel demand**: seasonal peaks in spring (planting) and autumn (harvest) create modest secondary demand bumps. USDA crop progress reports and planted acreage estimates indirectly affect the April–May and September–October distillate demand outlook.

4.2 The Diesel Fuel Quality Transition — ULSD and Its Impact

- **EPA ULSD mandate (effective June 2006 for on-road)**: EPA required all on-road diesel fuel to be ULSD (max 15 ppm sulphur) by June 2006, replacing Low Sulphur Diesel (LSD, 500 ppm) completely. Off-road diesel (construction, agriculture) followed with ULSD requirements phased in through 2010.
- The transition to ULSD required significant refinery investment in hydrotreating capacity across the industry. These capital costs are now sunk and embedded in the structural refinery economics of ULSD production.
- **IMO 2020 marine sulphur cap**: the International Maritime Organization's 2020 regulation reduced the maximum sulphur content of marine fuel from 3.5% to 0.5% globally (enforced January 1, 2020). This structural change increased demand for low-sulphur fuel oil (VLSFO) and marine gasoil (MGO), both of which draw from the distillate pool. The IMO 2020 transition was a structural bullish event for distillate demand and the HO forward curve — though the COVID demand collapse in 2020 masked much of the effect.

5. Supply — Refinery Economics and Distillate Yield

Distillate supply is determined by refinery runs, crude oil grade selection, and the yield of distillate from each barrel of crude processed. Unlike gasoline, which is predominantly produced via fluid catalytic cracking (FCC), distillate is produced in both straight-run distillation and via hydrocracking — giving refiners some ability to shift yield between gasoline and distillate at the margin.

5.1 Refinery Yield and the Gasoline-Distillate Switch

- **Crude distillation unit (CDU)**: separates crude into fractions by boiling point. Distillate (kerosene through gas oil range) is a straight-run product from the CDU. A typical US refinery produces approximately 20–35% distillate from a barrel of crude (depending on crude gravity and refinery configuration).
- **FCC vs hydrocracker yield competition**: FCC units maximise gasoline production from heavy gas oil feedstock. Hydrocracking units can be configured to produce more distillate. When HO crack spreads are wide relative to RBOB cracks, refiners with hydrocracking capacity can shift feedstock allocation toward the hydrocracker (more distillate, less gasoline). This "yield switch" is not instantaneous but occurs within a 2–4 week operational window.
- **Crude gravity and distillate yield**: lighter crude grades (higher API gravity) yield proportionally more naphtha and gasoline and less distillate. Heavier crude grades yield more residual fuel and distillate. When Gulf Coast refineries process more heavy sour crude (Mars, Arab Heavy, Canadian WCS), distillate yield per barrel of crude tends to increase. The WTI-WCS differential (light-heavy spread) therefore affects the distillate yield economics.
- **Winter operating mode**: refineries in the Northeast and Midwest adjust configuration in Q3–Q4 to maximise distillate output ahead of winter heating demand. This seasonal refinery mode shift is a supply-side factor that partially offsets the seasonal demand increase but typically does not fully compensate, creating the winter distillate tightness premium.

5.2 US Distillate Production and Refinery Landscape

The same PADD structure that governs RBOB supply applies to distillate, but with different flow patterns:

PADD	Region	Distillate Role	HO Curve Impact
PADD 1	East Coast	Very limited domestic refinery capacity. Primarily a demand region. Imports from Gulf Coast via Colonial Pipeline and by tanker from Europe (ARA). PADD 1 distillate stocks are the primary inventory signal for the HO prompt spread.	PADD 1 distillate stocks vs 5-year average is the most watched inventory metric for HO. Low PADD 1 stocks entering winter = bullish for Nov-Feb strip.
PADD 2	Midwest	Significant refining. Supplies Midwest diesel demand. Colonial Pipeline and other systems move excess to PADD 1 when economics support.	PADD 2 excess distillate flows to PADD 1 via Colonial. PADD 2 refinery maintenance affects supply availability to the East Coast.
PADD 3	Gulf Coast	Dominant production centre. Exports distillate to Latin America, Europe, and via Colonial to PADD 1. US Gulf Coast (USGC) distillate is a global swing supplier. Hurricane disruption is the primary supply risk.	Gulf Coast refinery utilisation is the primary production variable. Export economics (USGC vs ARA) determine how much stays domestic vs exports.
PADD 4	Rocky Mountain	Small, isolated. Minimal relevance for NYH.	Negligible direct impact.

PADD 5	West Coast / Alaska	Self-sufficient; isolated from East Coast. California CARB diesel specification limits imports/exports.	Minimal direct impact on NYH HO. California diesel supply issues affect CARBOB-adjacent pricing but not HO contract.

5.3 Gulf Coast Refinery Maintenance and Hurricane Risk

- **Spring turnaround (Feb–Apr)**: same refinery maintenance pattern as for RBOB, but the impact on distillate is somewhat different. The spring turnaround reduces distillate output during a period when heating oil demand is already declining — so the supply reduction is partially offset by the seasonal demand retreat. The net effect on distillate spreads is typically less extreme than for RBOB during the spring transition.
- **Autumn turnaround (Sep–Oct)**: for distillate, the autumn turnaround is more market-impactful than for gasoline because it occurs simultaneously with rising heating oil demand and the pre-winter stocking season. A refinery unit outage in September–October that reduces distillate output can accelerate PADD 1 stock draws and build the winter premium in the October–November–December HO strip.
- **Hurricane risk (June–November)**: same geography as RBOB — approximately 50% of US refining capacity is on the Gulf Coast. For distillate, a hurricane disruption creates both a production shortage and a diesel supply disruption for hurricane recovery efforts (emergency generators, construction equipment, logistics vehicles in the affected region). This "recovery demand" can sustain distillate demand for weeks after a major storm — longer than the direct refinery outage.

6. Inventory Data — The EIA Weekly Signal

The EIA Weekly Petroleum Status Report (published every Wednesday at 10:30 ET) contains the distillate inventory data that is the primary weekly fundamental driver of HO prompt spreads. The distillate components are more seasonal and more supply-constrained than gasoline, making the weekly report arguably more impactful for HO than for RBOB during the winter season.

Data Point	What It Measures	Key Comparison	HO Spread Implication
Total US distillate stocks (mb)	All distillate fuels (ULSD + LSD + heating oil) in primary storage, broken by PADD.	5-year average range (most important); year-ago; analyst consensus	Above 5-yr avg = bearish for winter strip. Below = bullish. PADD 1 breakdown is the critical regional component.
PADD 1 distillate stocks (mb)	East Coast distillate inventory. The most directly relevant number for NYH HO delivery.	5-year average range; year-ago; entering-winter threshold (~35–40 mb is considered low)	Entering winter (October) below 35 mb in PADD 1 with cold forecasts = acute near-term tightness risk for Jan-Feb contracts.
Distillate production (mb/d)	Domestic refinery output of distillate for the week.	Year-over-year; vs analyst estimate	Low production (maintenance, FCC reconfig) during pre-winter = accelerates PADD 1 draws; bullish for front HO.
Distillate imports (mb/d)	Volume of distillate imported, primarily to PADD 1 from Europe (ARA) and other sources.	ARA-NYH arb open/closed status; ARA distillate stocks level	Rising imports = ARA arb open; moderately bearish for prompt HO. Falling imports = arb closed; supply tightening.
Distillate exports (mb/d)	Volume exported (primarily from Gulf Coast to Latin America, Europe).	USGC vs ARA arbitrage; Latin American demand	High exports reduce domestic supply; bullish for domestic HO. Winter export demand from Europe is a significant variable.
Distillate product supplied (mb/d)	Implied demand = production + imports − exports − stock change.	Year-over-year; seasonal expectation	Most direct demand measure. Cold winter weeks will show 20–30% above normal product supplied.

6.1 PADD 1 Distillate Stocks — The Most Critical Inventory Metric

- PADD 1 distillate stocks entering the winter heating season (typically measured in early October) are the single most important structural variable for winter HO spread pricing. The historical reference levels:
- **Above 45 million barrels entering winter**: comfortable inventory. The winter premium in November through January HO contracts is compressed. Mild weather can be well-absorbed.
- **35–45 million barrels**: near-average. Weather is the dominant swing variable. A colder-than-average winter can tighten stocks; a mild winter maintains or builds them.
- **Below 35 million barrels entering winter**: tight inventory. The winter strip is bid up. Any sustained cold snap creates acute near-term shortage concerns. Historical winters with below-35mb PADD 1 stocks and cold weather (2004–05, 2013–14 Polar Vortex) produced extreme HO backwardation and price spikes.

The combination of PADD 1 stock level, temperature forecast, and the pace of Colonial Pipeline nominations defines the near-term supply adequacy for NYH heating oil delivery. When all three point negative simultaneously (low stocks + cold forecast + slow Colonial nominations), front HO spreads can move $0.10–0.30/gallon within days.

7. Global Trade Flows — The Atlantic Basin Distillate Market

Unlike most agricultural commodities, distillate markets are globally connected. The US Gulf Coast, the ARA (Amsterdam-Rotterdam-Antwerp) hub in Europe, the Arabian Gulf, and Singapore are the four primary global distillate hubs. Trade flows between them directly affect PADD 1 supply and therefore the HO forward curve.

7.1 The ARA-NYH Arbitrage

- **The distillate arb mechanics**: when NYH ULSD prices are sufficiently above ARA gasoil (adjusted for: tanker freight ~$0.03–0.08/gallon, quality adjustment, sulphur compliance cost, and transatlantic transit time ~7–10 days), European and Atlantic Basin product flows toward the US East Coast. This arbitrage naturally caps the premium of NYH ULSD over ARA gasoil in normal market conditions.
- **ARA distillate stocks (Insights Global / Genscape)**: published weekly. The level of ARA distillate stocks determines how much product is available for export to PADD 1. When ARA stocks are low (because European demand is high or Russian supply has been disrupted), less product is available for US import — tightening the domestic supply and supporting the HO spread.
- **Russian distillate — the historical supply pillar**: prior to 2022, Russia was the dominant supplier of distillate (gasoil) to the European market, providing approximately 1.0–1.5 mb/d of gasoil exports, primarily to European ports via tanker from Baltic and Black Sea terminals. Post-2022 sanctions and the G7 price cap on Russian petroleum products redirected Russian gasoil to India, Turkey, and other non-sanctioning nations and reduced European ARA gasoil supply. This structural supply loss was the primary driver of the exceptional HO and ICE Gasoil crack spread widening in 2022–23.
- **Middle Eastern distillate (Arabian Gulf)**: Saudi Arabia (Aramco), Kuwait Petroleum, and UAE refineries export distillate to Europe and Asia. When European supply is tight and the ARA-NYH arb is closed, Middle Eastern product can supply Europe while European barrels flow to the US — a "daisy chain" arbitrage. The Fujairah distillate stock report (weekly, UAE) is a proxy for Middle Eastern distillate availability.

7.2 ICE Gasoil — The European Benchmark and Its Relationship to HO

ICE Gasoil (ticker: GO, traded on ICE Futures Europe) is the European equivalent of NYMEX HO. It prices gasoil delivered at Rotterdam and is denominated in USD per metric tonne. The HO-Gasoil relationship is one of the most watched cross-market spreads in energy:

- **The HO-Gasoil "arb" or EFP**: the spread between NYH ULSD (in $/MT equivalent) and ICE Gasoil (in $/MT). When HO trades at a premium to Gasoil (the "arb is open"), European product flows west to the US. When HO trades at a discount to Gasoil (the "arb is closed"), US Gulf Coast product may flow east to Europe, or the two markets remain isolated.
- **The arb as a structural price ceiling for HO**: if HO rises far enough above Gasoil, transatlantic imports are economically attractive and flow, capping the HO premium. In practice, this cap is not perfectly elastic — it takes 7–10 days for product to transit, and if demand is rising rapidly (polar vortex event), imports may not arrive fast enough to prevent acute near-term price spikes.
- **Russian gasoil replacement post-2022**: with Russian gasoil displaced from the European market, both ARA stocks and the HO-Gasoil arb have structurally shifted. The European market now competes more directly with the US for Middle Eastern and Atlantic Basin distillate — a persistent structural tightening of the global distillate market.

7.3 The US as a Distillate Exporter

- The US Gulf Coast has become a significant net exporter of distillate, shipping approximately 1.0–1.5 mb/d of ULSD and gasoil to Latin America (Mexico, Brazil, Colombia, Chile, Argentina) and Europe.

Mexico is the single largest US distillate export destination.

- When the USGC-ARA arb is open (US product cheaper than European product), US Gulf Coast exports increase and domestic supply available for the Colonial pipeline and PADD 1 supply decreases. Tracking weekly EIA export data allows traders to assess how much supply is being diverted internationally.
- Latin American distillate demand is correlated with regional economic activity, agricultural seasons, and infrastructure investment cycles. It provides a relatively stable baseline export demand that grows gradually with regional GDP.

8. The Forward Curve Structure — Zones and Drivers

The HO forward curve has a characteristic shape dominated by the winter heating demand premium. Its structure is fundamentally different from RBOB: HO is peaked in winter, RBOB in summer. The seasonal relationship between the two drives the RBOB-HO spread, which is itself an important traded instrument.

Zone / Period	Structural Feature	Primary Driver	Typical Curve Shape	Key Signal
Nov–Feb (winter peak)	Peak heating oil demand. Coldest period. Minimum distillate production relative to demand.	PADD 1 stocks vs 5-yr avg; temperature (HDD); Colonial Pipeline throughput; ARA import arb status	Typically in backwardation relative to spring months. The size of backwardation = market's assessment of winter tightness.	EIA weekly PADD 1 stocks; NOAA 6–10 day temp outlook; ARA stock level
Mar–Apr (end of winter)	Heating demand declining. Demand transition from heating oil to diesel dominance.	End-of-winter PADD 1 stock trajectory; refinery maintenance reducing distillate output	Feb-March HO spread can be in backwardation if winter was cold and stocks are low entering March.	Final winter HDD accumulation; spring turnaround schedule for Gulf Coast distillate output
May–Sep (summer / highway diesel)	Heating demand near zero. Highway diesel dominates. Agricultural diesel demand peaks in Apr-May and Sep-Oct.	GDP / freight (ATA Truck Tonnage); ISM Manufacturing; export arb economics (USGC vs ARA); refinery utilisation	Mild contango typical. The summer HO is priced around highway diesel demand and carry economics.	EIA product supplied (highway diesel); ATA Truck Tonnage Index; ISM; USGC export pace
Oct–Nov (pre-winter build)	Distillate pre-buying season. Heating oil dealers stock up. Refinery mode shifting toward distillate maximisation.	PADD 1 stock build pace; temperature forecasts for winter; CPC 90-day outlook; autumn refinery maintenance	October-November spread historically widens as market builds in winter premium. The size of the Oct-Nov spread is the clearest early-season winter tightness signal.	Oct 31 PADD 1 stock level (the key seasonal reference point); CPC winter temperature outlook
Mid-curve (C6–C24)	Structural diesel demand outlook. Energy transition pace.	GDP trend; freight demand growth; electric truck adoption curve; IMO regulatory evolution	Mild contango reflecting cost of carry and structural demand growth assumptions.	EIA AEO; IEA long-term demand outlook; electric truck adoption data

8.1 The RBOB-HO Seasonal Spread

The spread between RBOB and HO (both quoted in $/gallon at NYH) is one of the most actively traded refined product spreads:

- **Summer: RBOB at premium to HO**. Driving season demand supports gasoline. Heating oil demand near zero. Refinery yield shifts toward gasoline maximisation. Typical summer RBOB premium: $0.05–$0.15/gallon.
- **Winter: HO at premium to RBOB**. Heating demand drives distillate. Cold weather events spike HO relative to RBOB. Refinery yield shifts toward distillate maximisation. Typical winter HO premium: $0.05–$0.20/gallon; can spike to $0.50+/gallon in extreme cold.
- **The transition months (March-April, September-October)**: the seasonal relationship between RBOB and HO reverses twice a year. The timing and magnitude of these reversals depend on the relative tightness of each product market and the balance of refinery yield between them. These transition periods are the most volatile for the RBOB-HO spread.

- **The RBOB-HO spread as a refinery yield signal**: when RBOB is at a large premium to HO, refineries are incentivised to maximise gasoline yield (shift FCC feedstock, reduce hydrocracker distillate). When HO is at a large premium, the opposite yield shift occurs. This self-correcting mechanism moderates extreme deviations in the RBOB-HO spread over a 2–4 week operational horizon.

9. Weather Data for Distillate Markets

Weather affects HO demand primarily through heating degree days (HDD) in winter, with a secondary effect from cold weather impacts on agricultural logistics. The weather data inputs for HO are similar to but distinct from those for natural gas (see the Natural Gas reference document):

- **Population-weighted HDD for the Northeast US**: the most relevant weather metric for heating oil demand. Unlike gas (which is consumed nationwide), heating oil demand is geographically concentrated in New England and New York. The per-HDD heating oil consumption is higher in the Northeast than any other region. NOAA and commercial weather services (DTN, WeatherBell, Maxar) provide daily Northeast HDD actuals and forecasts.
- **The ECMWF model for 6–15 day temperature forecasts**: same reference model used for natural gas. A shift in ECMWF ensemble temperature forecasts for the US Northeast by 5°F for a week is equivalent to a change of ~35 HDD and represents a meaningful heating demand change. Large model forecast shifts move HO futures immediately.
- **The Arctic Oscillation (AO) and North Atlantic Oscillation (NAO)**: two atmospheric circulation patterns that strongly influence Northeast US winter temperatures. A negative AO/NAO index is associated with cold Arctic air outbreaks into the eastern US — the precursor to polar vortex events. Meteorologists track the AO/NAO index as a 7–14 day leading indicator of severe cold risk in the Northeast. A shift to strongly negative AO/NAO is an acute bullish signal for the nearest HO contract.
- **Seasonal ENSO effects on distillate demand**: El Nino winters in the US tend to be warmer-than-average in the northern tier and Northeast, bearish for heating oil demand. La Nina winters tend to be colder in the north, bullish. These signals affect the pricing of the November–February HO strip starting in September–October as forecasters publish their winter outlooks.
- **The CPC 90-day seasonal temperature outlook**: published monthly by NOAA's Climate Prediction Center. The October update is the most market-critical publication of the year for the winter HO strip: a cold (below-normal probability) forecast for November–January issued in October immediately creates a bid for the winter contracts.

10. Macro, Regulatory, and Cross-Market Drivers

10.1 The HO-WTI Crack Spread and Refinery Economics

- **HO crack spread seasonality**: the HO-WTI crack spread is highest in winter (when heating oil demand peaks) and lowest in summer. This is the mirror image of the RBOB crack seasonal pattern. In practice, the combined 3-2-1 crack spread (2 RBOB + 1 HO − 3 WTI) has a smoother seasonal pattern because the RBOB summer premium and HO winter premium partially offset each other.
- **Crude grade impact on distillate yield**: sour crude grades (Arab Medium, Arab Heavy, Iraqi Basra Heavy, WCS) yield proportionally more distillate than light sweet grades. When heavy sour crude prices fall relative to light sweet (wide light-heavy differential), complex refineries processing more heavy crude earn better distillate margins. Gulf Coast complex refineries respond by increasing heavy crude intake when the WTI-WCS differential is wide.
- **Hydrocracker economics**: refineries with hydrocrackers can flex distillate yield within approximately 5–10% of crude input. Wide HO-RBOB crack spreads (HO at premium) incentivise shifting feedstock from FCC (gasoline) to hydrocracker (distillate) configuration. This yield response typically unfolds over 2–4 weeks.

10.2 IMO 2020 — Structural Impact on Distillate Markets

The IMO 2020 global marine sulphur cap (implemented January 1, 2020) was the most significant structural change to distillate markets since the US ULSD mandate:

- **What changed**: ships were required to use fuel with ≤0.5% sulphur (down from 3.5%) globally. Ships could either switch to VLSFO (Very Low Sulphur Fuel Oil, a blend of fuel oil and distillate), to MGO (Marine Gas Oil, essentially ULSD), or install exhaust gas scrubbers.
- **Impact on distillate demand**: approximately 250,000–350,000 b/d of incremental global distillate demand was created by IMO 2020 (from ships switching to MGO and from distillate blending into VLSFO). This structural demand increase shifted the global distillate supply/demand balance and added a permanent floor under the gasoil/distillate crack spread relative to pre-2020 norms.
- **Scrubber adoption**: approximately 15–20% of the global commercial fleet by tonnage has installed exhaust gas scrubbers, allowing continued use of high-sulphur fuel oil (HSFO). This segment is not distillate demand, but it competes with VLSFO users for the cost advantage. When HSFO is significantly cheaper than VLSFO, the economics of scrubber payback improve, potentially limiting future MGO/distillate demand growth.

10.3 Diesel-Gas Switching in Power and Industry

- In industrial settings with dual-fuel capability (large generators, industrial boilers), when natural gas prices spike, some demand shifts to diesel/HO. During the Polar Vortex events (2014, 2019) and during the European gas crisis (2021–22), this switching added incremental distillate demand on top of already-stressed inventories.
- Power generation using diesel oil (primarily emergency backup generators) is typically small in total volume but can be acute in regional markets (New England power plants burning heating oil when gas prices spike) — an acute local bullish signal for PADD 1 distillate.

10.4 Speculative Positioning and COT

- **CFTC COT for HO**: weekly managed money net position. HO managed money positioning builds long heading into winter (September–November) as hedge funds position for the seasonal demand increase. The unwinding of this positioning in Q1 (as winter demand recedes) contributes to the seasonal weakening of the winter premium.
- **Commercial short (producer/hedger) position**: heating oil distributors and distillate suppliers hedge forward purchases by buying HO futures (establishing long commercial positions), while

refiners hedge forward production by selling. The balance of these commercial positions provides a signal about the market's commercial supply-demand view.

- **Diesel price caps and policy risk**: several countries have implemented temporary diesel price caps or export restrictions during periods of high fuel prices (India's diesel export tax in 2022, Argentina's petroleum product price controls). These policy interventions can disrupt normal trade flow patterns and affect the availability of distillate in the Atlantic Basin.

11. Documented Seasonal Patterns

Heating Oil / ULSD has highly seasonal patterns driven by the dual demand structure — winter heating and year-round highway diesel. The seasonal patterns are among the most consistent in petroleum markets but are subject to significant inter-year variation based on temperature.

Period	Heating Demand	Diesel Demand	Supply Factor	Typical Spread Direction	Reliability
Oct–Nov (pre-winter build)	Pre-buying begins. Dealers stock. HDD small but rising.	Autumn harvest; freight steady.	Autumn turnaround partially reduces refinery distillate output.	Oct-Nov spread widens (backwardation building). The pace of widening signals market's winter tightness assessment.	High
Dec–Jan (winter peak)	Maximum seasonal demand. HDD at annual high.	Year-round steady freight.	Refinery running at full distillate mode. Pipeline and import maximum.	Near-term backwardation at maximum. Front contracts spike on cold events. Jan-Feb spread most volatile.	High — but magnitude varies widely with temperature
Feb–Mar (late winter)	Demand declining but still meaningful. Late cold snaps (polar vortex risk).	Steady. Freight recovering from post-holiday dip.	Spring turnaround beginning at some refineries.	Feb-March spread can remain in backwardation if stocks are low. Transition from backwardation to contango occurs in March-April typically.	Moderate — late cold snaps are significant wildcards
Apr–May (spring)	Near zero. Heating season over.	Agricultural planting peak. Freight recovering. Highway diesel steady.	Refinery maintenance peak reducing distillate output; offsets demand drop partially.	Mild contango. The RBOB-HO spread reverses (RBOB moves to premium as driving season approaches).	High
Jun–Sep (summer)	Zero heating demand.	Summer freight (high); agricultural harvest approaching in Sep.	Full refinery utilisation; maximum distillate production.	Mild contango. HO at seasonal discount to RBOB. Hurricane risk adds premium to late-summer.	Moderate (hurricane risk)
Sep–Oct (shoulder)	Minimal but beginning to tick up late in period.	Harvest peak. Year-round freight.	Autumn turnaround at some refineries reduces output.	Transition period. The Oct-Nov spread begins to reflect winter premium. Oct 31 PADD 1 stocks set the seasonal starting point for winter.	High — the Oct 31 stock level is the most important annual data point for winter HO

11.1 Key Documented Historical Events

- **Polar Vortex — January 2014**: extreme cold across the Northeast and Midwest. PADD 1 distillate stocks had entered winter at relatively low levels. HO front-month spiked over $0.40/gallon in one week. The January-February HO spread went into extreme backwardation. Algonquin natural gas basis simultaneously spiked (demonstrating the dual-fuel switching dynamic).
- **Russian gasoil sanctions (2022–2023)**: EU ban on Russian petroleum product imports (implemented February 2023) removed approximately 0.7–1.0 mb/d of gasoil from the European market. ARA distillate stocks fell sharply; the ARA-NYH arb inverted (US barrels flowing east to Europe rather than west from Europe to the US). ICE Gasoil and NYMEX HO crack spreads reached multi-year highs. US Gulf Coast distillate exports to Europe increased.
- **Hurricane Harvey — August 2017**: Gulf Coast refinery shutdowns created both a crude oversupply (refineries not buying) and a product shortage (no production). For distillate specifically, Harvey recovery demand (generators, construction equipment, trucks) sustained the bullish pressure for weeks after refinery restarts.
- **COVID-19 demand collapse — Spring 2020**: road transport and aviation demand collapsed simultaneously. Distillate product supplied fell ~25% year-over-year in April 2020. PADD 1 stocks built rapidly. HO crack spreads compressed to historic lows. The IMO 2020 structural demand increase was completely masked by the COVID demand destruction.

12. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Release Timing	Primary HO Curve Impact
EIA Weekly Petroleum Status Report — Distillate stocks (PADD 1 critical), production, imports, exports, product supplied	US EIA	Weekly	Wed 10:30 ET	Prompt C1–C3 (winter season); year-round for highway diesel signal. PADD 1 distillate most watched number.
EIA Short-Term Energy Outlook (STEO)	US EIA	Monthly	2nd Tue of month	C2–C12 — distillate supply/demand balance; refinery utilisation; demand forecast revision
NOAA CPC 6–10 / 8–14 day temperature outlook	NOAA CPC	Daily / Twice weekly	Continuous	C1–C2 (winter) — Northeast temperature forecast; primary weather-driven HO signal
NOAA CPC monthly temperature outlook	NOAA CPC	Monthly	~15th of month	C2–C3 (winter) — medium-range temperature probability for Northeast heating demand
NOAA CPC 90-day seasonal temperature outlook	NOAA CPC	Monthly	~15th of month	C3–C5 (Oct outlook is most critical for winter strip pricing)
Arctic Oscillation (AO) / North Atlantic Oscillation (NAO) index	NOAA CPC	Daily	Continuous	C1–C2 (winter) — 7–14 day leading indicator of severe cold risk in the Northeast
ENSO forecast update	NOAA CPC / IRI	Monthly	~Mid-month	C4–C7 (winter strip) — seasonal temperature signal for winter heating demand outlook
Insights Global ARA distillate stocks	Insights Global	Weekly	Wed–Thu	C1–C3 — European distillate supply available for NYH import; ARA-NYH arb status
ICE Gasoil futures (GO) price and structure	ICE Futures Europe	Continuous	Trading hours	C1–C6 — HO-Gasoil spread (the arb); European distillate price reference
Fujairah distillate stock report	Fujairah Oil Industry Zone	Weekly	Monday–Tuesday	C2–C4 — Middle Eastern distillate availability; supplement to ARA signal
ATA Truck Tonnage Index	American Trucking Assoc.	Monthly	~3rd week of month	C2–C6 — highway diesel demand proxy; freight activity signal
Cass Freight Index	Cass Information Systems	Monthly	~3rd week of month	C2–C6 — freight shipment volumes; leading indicator of diesel demand
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET	Positioning / crowding — all months; winter buildup of managed money longs most relevant
Baker Hughes Rig Count	Baker Hughes	Weekly	Fri ~13:00 ET	C3–C12 — crude production indicator; feedstock for Gulf Coast distillate production
EIA Annual Energy Outlook	US EIA	Annual	Jan–Feb	C24+ — long-run diesel demand trajectory; electric truck adoption; IMO evolution
IMO regulatory updates (MEPC meetings)	International Maritime Organization	Biannual (approx.)	Variable	C12–C36 — future marine sulphur and carbon regulations affecting marine distillate demand

NYMEX Heating Oil / ULSD occupies a unique position in the petroleum product landscape: it is simultaneously a seasonal weather-driven commodity (heating oil demand), a cyclical freight and industrial commodity (highway diesel), a globally traded product (connected to ARA, Arabian Gulf, and Singapore via trade arb), and a regulated marine fuel (post-IMO 2020 marine gasoil and VLSFO demand). No other petroleum product futures contract carries this breadth of fundamental drivers across its curve.

> ***For spread traders, the most reliable edges come from three recurring patterns: (1) the October-November pre-winter stock build trade — when PADD 1 distillate inventories enter October below the 5-year average, the winter strip is structurally underpriced relative to historical norms and cold forecasts create convex upside; (2) the end-of-winter (February-March) stock draw rate — the pace of PADD 1 draws relative to the remaining winter determines whether winter backwardation persists into March or collapses rapidly; (3) the ARA-NYH arbitrage signal — when the transatlantic arb is open and ARA stocks are high, imports will flow and the winter premium compresses; when the arb is closed and ARA stocks are low (as occurred post-2022 with Russian supply disruption), the winter premium can sustain at historically elevated levels.***